Peeyush Khare, Ph.D.

Visit lab group website: Click here - Khare Research Lab

Research interests: Environmental impacts of anthropogenic activities in megacities; Emerging pollutants in future sustainable cities; Influence of unregulated-burning activities on urban air quality; Development of analytical techniques and automation software for environmental research

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Contact Information

104 Center for Environmental Science and Engineering
Kotak School of Sustainability
Indian Institute of Technology (IIT), Kanpur, India

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peeyushkhare90@gmail.com

Academic Appointments

Assistant Professor (Sept. 2026-current)	**Indian Institute of Technology (IIT), Kanpur, India** Kotak School of Sustainability
Research Scientist TVöD E14L4 (June 2024-Aug 2026)	**Forschungszentrum Jülich, (Helmholtz-Gemeinschaft) Germany** Institute of Climate and Energy Systems (ICE-3) - Troposphere Reactive Trace Substances Division
Marie S.-Curie CoFund Postdoctoral Fellow (Dec 2020-May 2024)	**Paul Scherrer Institute (PSI), Switzerland** PSI Center for Energy and Environmental Sciences Laboratory of Atmospheric Chemistry

Education

Ph.D. (2020) Ph.M. (2018) M.S. (2017)	**Yale University, USA** Department of Chemical and Environmental Engineering **Honors in 8 of 9 graduate courses**.
PGDip (2015)	**Sciences Po Paris France - Ashoka University India (Joint Program)** Environmental Policy (PSIA, SciencesPo) and Liberal Arts (YIF, Ashoka)
M.S. (2014)	**Virginia Tech, USA** Department of Civil Engineering (Environmental)
B.Tech. (2012) (Accredited by ABET, USA)	**Vellore Institute of Technology, Vellore India** Department of Civil Engineering **GPA: 9.33/10.0** (Highlight: ***Best Outgoing Student***; *Top-10 university-wide*)

Research Theses

Ph.D. (Yale University)	*Understanding the Increasing Influence of Non-Combustion Sources on Urban Air Quality.* Advisor: Prof. Drew R. Gentner (Chemical and Environmental Engineering)
M.S. (Virginia Tech)	*Simulation of Vertical Conc. Gradient of Influenza Virus Resuspended from Floor Dust* Advisor: Prof. Linsey Marr (Civil Engineering)
B.Tech. (VIT Vellore)	*Tackling Failed Monsoons: A Sustainable Approach* Advisor: Prof. Sat Ghosh, FRMetS (Civil Engineering)

Professional Experience Highlights

Forschungszentrum Jülich (June 2024-Aug 2026)

Collaborators: ACTRIS National Facilities, IONICON Analytik Austria, TOFWERK Switzerland, University College Cork Ireland, NOAA USA

- P.I. for Fusion-ToF measurements in international chamber campaigns.
- Ambient measurements using high-resolution Time-of-Flight mass spectrometry.
- Development of ACTRIS chamber measurement guidelines.
- Contributions to ACTRIS PTR-Time-of-Flight measurement guidelines.
- Pan-European audits of ACTRIS measurement sites.
- Advising Ph.D. students and postdocs.

Paul Scherrer Institute (Dec. 2020-May 2024)

Funded by: Swiss Agency for Development and Cooperation (SDC) and the European Commission Horizon 2020 Marie Curie COFUND Grant

Collaborators: TERI, IIT-Kanpur, IIT-Delhi, NEERI, University of Berne, University of Grenoble, Spanish National Research Council.

- Characterization, sources and oxidative potential of ambient PM2.5 in India.
- Source apportionment for ambient PM2.5 in Barcelona, Spain.
- Developing novel techniques for chemical characterization of organic aerosol.

Supporting contributions to:

- Mobile measurements in Sarajevo, Bosnia using HR-ToF-AMS and Vocus PTR-ToF.
- Offline characterization of PM2.5 from megacities in China and Switzerland.
- Development of offline EESI-Orbitrap instrumentation techniques.

Yale University (2015 - 2020)

Funded by: Yale Fellowship & grants from the National Science Foundation (NSF), U.S. Environmental Protection Agency (EPA), NYSERDA and NESCAUM.

Virginia Tech (2012 - 2014)

Funded by: Institute of Critical Technology and Applied Science

Larsen & Toubro, ECC (2010)

VIT Vellore (2009-2012)

Funded by: Villgro Possible and VIT Patent Office

Collaborators: Aerodyne Research, Carnegie Mellon University, U.S. EPA

- Modeling non-combustion emissions in Greater Los Angeles, California.
- Developing techniques to characterize unresolved complex mixture emissions.
- Designing experiments to study environmental impacts of asphalt materials.
- NH4+-ToF field measurements in wintertime New York City on urban emissions.
- Oxidation flow reactor experiments at Aerodyne Research to study asphalt SOA.
- ***(solo, PhD-unrelated):*** Ground-water analysis in 6 districts of central India.
- Sorbent tubes sampling campaigns in New York City (2017, 2018).
- Provided I/SVOC emissions data from consumer products to U.S. EPA inventories.

- Experiments to measure turbulent eddies during indoor movement, and;
- FORTRAN-based mathematical modeling to simulate vertical transport of influenza virus indoors resuspended with floor dust during human movement.

Supporting contributions to:

- Vision-based monitoring of the carbon footprint of construction equipment.

Summer Intern, Shrinagar Hydroelectric Power Project, Uttarakhand India

- Training: Reinforced concrete design of the dam, turbine housing & draft tube.

- Fabricated a novel rainfall simulator suitable for hydrological studies in India.
- Stochastic analysis of released droplet trajectories in FORTRAN.
- Received patent from Government of India for the instrument design.

Funded Research Activities

PI: Offline Mass-Spectrometric Analyses of Ambient Particulate Matter Filters from India
Source: The European Commission, Horizon 2020 Marie S. Curie CoFund program
EU Grant Agreement No. 884104
Budget: approx. USD 100,000. (One hundred thousand U.S. Dollars)

Co-PI: Development of Fixed-Wing Drone Platform for Atmospheric VOC Monitoring
At Forschungszentrum Jülich within the ACTRIS CiGas Framework
ACTRIS funding sources: German Federal Ministries (BMBF, BMUV) and institute resources at Forschungszentrum Jülich

Other Grants

- Full Sponsorship, Wilhelm and Else (WE) Heraeus-Foundation Seminar Germany, 2023. (USD 1500)
- Yale Graduate Student Assembly Conference Travel Award, 2019. (USD 750)
- American Geophysical Union: Fall Meeting Travel Grant, 2017. (USD 500)
- NASA Travel Grant for American Association for Aerosol Research Conference, 2017. (USD 500)
- Full Sponsorship, International Students Committee, St. Gallen Symposium, 2016. (USD 2000)
- American Meteorological Society 94th Annual Meeting Travel Grant, 2014. (USD 500)
- Villgro Possible, India- Young Innovator Funding Award, 2011. (USD 500)

Patents

Government of India, The Patent Office. Patent no. 351688 on application: 3623/CHE/2011
Name: *Varun: A Unique Rain Simulator*
Inventors: **Peeyush Khare**, Tushar Agarwal and Satyajit Ghosh
Patentee: Vellore Institute of Technology, VIT University Vellore T.N. India
Purpose: A new rainfall simulator that mimics Indian natural rainfall cost effectively.

Select Awards/ Honors

- **Young Excellent Scientist Program 2025 Awardee** at Forschungszentrum Jülich.
  §Nominated among 6 scientists from across all disciplines of the research center.
- **Assistant Professor position** offered at the University of Arizona, USA 2023. (declined)
  §Competitive startup package of USD 1.34 million with a tenure track position.
- **Full member, Sigma Xi** - The Scientific Research Honor Society, USA
  §Only via nomination.
- **Winning Pitch** at the Hackathon by TUMCREATE and Singapore-ETH Centre at the GYSS 2024, Singapore.
  §Proposal: SIMPLE - System for Intelligent Management of Pollutants and Emissions.
- **Atmos. Chem. Colloquium (ACCESS XVI)** program attendee, Brookhaven National Laboratory 2021, USA.
- **Marie S.-Curie CoFund Fellowship** by the European Commission for scientific research at PSI, 2021, Switzerland.
- **St. Gallen Wings of Excellence Award**: Leader of Tomorrow, St. Gallen Symposium 2016, Switzerland.
  §Paper chosen in top-21 articles among ~1000 students from 108 nations/350 universities.
- **Young India Fellow** 2014-15.

- **Best Outgoing Student** (School of Mechanical and Building Sciences: Civil+Mech.+Chem. Engineering)
- **Top-10 university-wide in the 2012 graduating batch of ~4000 students** (VIT Vellore), 2012.
- Industrial Endowment Award for excellence in academics, VIT Vellore, 2012**.**
- Certificate of Distinguished Research Achievements, VIT School of Mechanical and Building Sciences, 2012.
- **Rank 2/72** in the B.Tech Civil Engineering Batch of 2012, VIT Vellore.

Select Academic Scholarships/ Fellowships

- Yale Teaching Fellowships in:

ENAS 335 Professional Ethics, Fall 2018 and 2019.
ENAS 673/ENVE 473 Air Quality and Energy, 2018 and 2019.
PHY 180 Fundamentals of Physics, 2018.
ENAS 120/ENVE 120 Introduction to Environmental Engineering, 2017

- Yale Graduate Fellowship 2015-16.
- Charpak Scholarship (Government of France), 2015.
- Sciences Po Paris full tuition waiver, 2014.
- Balkrishna Industries Tyres Limited (BKT) Scholarship for graduate studies at YIF, 2014.
- Virginia Tech Research and Teaching Assistantships 2012-2014.
- Virginia Tech full tuition waiver for M.S. studies, 2012-2014.
- VIT Annual Merit Scholarships **(All four undergraduate years)**, 2008-2012.

Competitive Invitations to Special International Summits/Conferences

- Invited to the **Molecular-Level Understanding of Atmospheric Aerosols (MUOAA) 2024**, Corsica, France.
- Invited to the **Global Young Scientists Summit 2024** by N.R.F, Prime Minister's Office Singapore.
- Delegate at the **4th European Union Clean Air Forum** by the European Commission, Rotterdam 2023.
- Invited to the **782. WE Heraeus Seminar 2023** in Aerosols, Health and Climate: Gigacities and Future, Germany.
- Invited to the **Gordon Research Conference (GRC) in Atmospheric Chemistry 2019**, Newry Maine USA.
- Environmental Scientist delegate at the **World Bank Youth Summit 2018**, W.B. Headquarters Washington DC USA.
- Environmental Scientist delegate at The **Youth Assembly 2018**, NYC USA.

Licenses and Certifications

- *Professional Engineer In-Training (EIT)- Environmental Engineering*

Connecticut Board of Examiners for Professional Engineers and Land Surveyors USA
Verifiable link: https://account.ncees.org/rn/1825164-1026646-7d786e0

- *Data Analysis with Python (IBM)*

Coursera (Issued May 2023). Grade: 93%
Credential ID 3A2BJHN838SH
Link to certificate: https://coursera.org/share/b9f93a83486ddb16d964fc13e425e23b
Skills: Predictive modeling, Model selection, Data Analysis and Visualization

Laboratory Instrumentation Skills

At the Institute of Climate and Energy Systems (ICE-3): Troposphere, Forschungszentrum Jülich:

- CHARON Fusion-time-of-flight mass spectrometry (IONICON)
- Working with SAPHIR environmental chamber and CACTUS flow tube.
- IoniToF and Ionicon Data Analyzer (IDA) toolkits for ToF data analyses.

At Paul Scherrer Institute, Switzerland:

- High resolution time-of-flight aerosol mass spectrometry (LTOF- and HTOF-AMS)
- Extractive electrospray ionization-Ultra high-resolution orbitrap mass spectrometry (EESI-Orbitrap MS)
- EESI-LTOF MS; EESI-HTOF MS
- In-lab aerosol generation techniques (ESI autosampler and nebulizer)
- Total organic carbon analyzer
- Scanning mobility particle sizer (SMPS)
- Developed analytical methods for EESI-based measurements of non-water-extractable ambient organic aerosol composition

At Gentner Lab, Yale:

- Thermal desorption coupled with gas chromatography electron ionization-mass spectrometry (TD GC EI-MS)
- TD GC-atmospheric pressure chemical ionization-quadruple-time of flight mass spectrometry (TD GC APCI-Q-TOF)
- Liquid chromatography-electrospray ionzation-Q-TOF (LC ESI-Q-TOF)
- Thermal desorption and thermal conditioning techniques for adsorbent quality control.

In collaboration at Aerodyne Research, Billerica MA:

Conducted experiments using potential aerosol mass (PAM) reactor coupled with:

- Time-of flight aerosol mass spectrometer (ToF-AMS) [IGOR toolkits used: Squirrel and Pika]
- VOCUS proton transfer reaction mass spectrometer (VOCUS PTR-TOF) [IGOR toolkits used: Tofware]
- VOCUS ammonium time-of-flight mass spectrometry (VOCUS NH4+-TOF) for urban ambient measurements

Programming Skills: Igor Pro, R, Python, Interactive Data Language, Fortran.
Software Developed
<In Python> MOLINA, to automate TOF data analysis for IONICON Fusion-TOF instruments.
<In IGOR Pro> TURACO, to process PIKA and TOFWARE outputs for PMF in SoFi.
<In IGOR Pro> CHETAK, to automate the analysis of high resolution GC APCI-Q-TOF data.
Demonstration: Click for Khare video (Auto calibration: 21:00, Auto speciation: 45:00)
Languages : English/Hindi: Native proficiency, French/German/Spanish: Basic/Learner

Invited Talks at:
[14] Invited Panelist Speaker, Higher Education Series: Panel on Environment, Ashoka University, India, July 2026
[13] Future of Chemistry Symposium, Dept. of Chemical Sciences, Tata Institute of Fundamental Research, Mumbai 2026
[12] Environmental Science and Engineering Department, Indian Institute of Technology (IIT), Bombay 2026
[11] Department of Civil Engineering, Indian Institute of Technology (IIT), Kanpur 2026
[10] Science Online Talk, Young Excellent Scientist Program, Forschungszentrum Jülich, Germany 2025
[9] Center for Atmospheric Science, Cambridge University, United Kingdom 2023
[8] Department of Chemistry, Lehman College, City University of New York, USA 2023
[7] Distinguished Seminar Series, Department of Civil and Environmental Engineering, Northeastern University, USA 2023
[6] Department of Chemical and Environmental Engineering, University of Arizona, USA 2023
[5] School of Marine and Atmospheric Sciences, State University of New York (SUNY) at Stony Brook, USA 2023
[4] Institute of Energy and Climate Research, Troposphere, Forschungszentrum Jülich, Germany 2022, 2023
[3] EMPA Laboratory for Air Pollution and Environmental Technology, Switzerland 2022
[2] Atmospheric Chemistry Department, Leibniz Institute for Tropospheric Research (TROPOS), Germany 2022
[1] Laboratory of Atmospheric Chemistry, Paul Scherrer Institute, Switzerland 2020

Mentorship Experience

LAC graduate students Paul Scherrer Institute (2021)	Mentees: Jens Top, Marion Gosselin Research: *High-resolution chemical speciation of airborne particulate matter from Barcelona, Xi'an (China), Delhi and Lucknow (India).*
Teaching Fellow Yale University Fall (2018 and 2019)	ENAS 335 Professional Ethics Course instructor: Prof. Mercedes Carreras Duties: Graded assignments and helped organize guest lectures.
Teaching Fellow Yale University Spring (2018 and 2019)	ENAS 673/ENVE 473: Air Quality and Energy Course instructor: Prof. Drew R. Gentner Duties: Graded assignments.
Mentor Undergraduate Research Yale University (2018)	Mentee: Ricardo Soto Research Title: *An Examination of Volatile Organic Compound Emissions from Consumer and Industrial Products* Science, Technology and Research Scholars (STARS) Programme
Summer-Term Teaching Yale University (2018)	PHY 180 Fundamentals of Physics Course instructor: Dr. Mehdi Ghiassi-Nejad Duties: Held discussion hours, graded assignments and exams.
Teaching Fellow Yale University Fall (2017)	ENAS 120/ENVE 120: Intro to Environmental Engineering Course instructor: Prof. Jordan Peccia Duties: Held discussion classes and graded assignments.
Graduate Teaching Assistant Virginia Tech (2012-2014)	CEE 4324: Open Channel Flow CEE 5324: Advanced Hydrology CEE 5734: Urban Hydrology Course instructor: Prof. Glenn Moglen Duties: Graded assignments and proctored examinations.

Leadership and Service

ACTRIS auditor

Responsible for auditing ACTRIS CiGas atmospheric measurement stations pan-Europe:
- Jungfraujoch research station, Switzerland (November 2024)
- PALLAS-Sodankylä research station, Finland (March 2025)
- Hyytiälä research station, Finland (March 2025)
- SIRTA research station, France (December 2025)

Institute service

- Co-Organizer, ACTRIS CiGas workshops at **ACTRIS Weeks.**
Including: 2024, Italy; **2025**, Portugal
- 2022, *Lead*, Source apportionment bi-weekly meetings at the Lab. Atmos. Chem., PSI
- *2017, 2018 Member*, Graduate Student Recruitment Committee, Yale University
- 2018 *Member*, The Graduate Environmental Engineering Committee, Yale University
- 2019, *Organizing committee*, The Langer Symposium of the Yale Chemical and Environmental Engineering

Non-peer-reviewed publications on mentorship

- *Interactive Learning is the Only Way Out.*
Brainfeed Magazine India, vol. V, issue 5, August 2017.
- *High School to Engineering College.*
Brainfeed Magazine India, vol. IV, issue 8, November 2016.
- *Orienting Young Minds Towards Research and Its Implications for National Prosperity.*
Brainfeed Magazine India, vol. IV, issue 7, October 2016.
- *Can Technology Substitute the Teacher?*
Brainfeed Magazine India, vol. IV, issue 6, September 2016.

Other non-peer-reviewed publications

- *Smart Rain Simulators for Rain-Starved Districts of the Indian Subcontinent*
Khare, P., Agarwal, T., Ghosh, S.
Earth and Space Science Open Archive (ESSOAr)
https://doi.org/10.1002/essoar.10507296.1.
- *Transcending Boundaries: My Journey from Engineering to Social Sciences and Back.*
New Delhi Times, India
Published as headlines news in the online edition on August 28, 2019.
Link: https://www.newdelhitimes.com/transcending-boundaries-my-journey-from-engineering-to-social-sciences-and-back/
- Contributed a featured article to the book *"A Primer for How to Prepare for Overseas Postgraduate Admissions in Engineering"*, Macmillan Education Publications India.
Authored by: Prof. Sat Ghosh and Ms. Usha Jedusan
Affiliated with: Vellore Institute of Technology, VIT University India

Service to academic/scientific community

- **Reviewer**, Nature Geoscience (Nature); Science Advances (Science); Environmental Science and Technology Letters (ACS); Journal of Geophysical Research: Atmosphere (AGU); Environmental Science: Processes & Impacts Journal (RSC); Atmospheric Chemistry and Physics (EGU); Sustainability Chemistry & Engineering (ACS); Aerosol Measurement Techniques (EGU); Geophysical Research Letters (AGU); Journal of Analytical Atomic Spectrometry (RSC); Atmospheric Science Letters (RMetS); Journal of Materials in Civil Engineering (ASCE); Urban Climate (Elsevier)
- **Proposals Reviewer** for the United States Department of Energy SBIR programs

Public outreach

- *The Curious Young Mind: YIF Alumnus' Journey from Engineering to Liberal Arts to Bringing Environmental Change Worldwide.*
Interviewed and published in Ashoka University bulletin (and other social media handles) by Shreya Chatterjee (Ashoka University). December 5, 2020
- Interviewed for the special feature section of the Indian national magazine 'Brainfeed' in the article titled 'A Young India Fellow at Yale', vol. IV, issue 4, July 2, 2016.
- Interviewed for the education edition of the Indian national newspaper 'The Hindu' in the article titled 'A Young India Fellow at Yale' on May 30, 2016.
- Interviewed for the education edition of the Indian national newspaper 'The Times of India' in the article titled 'Clear The Air' on May 23, 2016.
- Cover page interview for the Indian national newspaper 'The Indian Express's Edex (education edition)' in the article titled 'Branch Out Expand Your Boundaries' on January 12, 2015.

Other Achievements

- Falling Walls Lab, Switzerland 2022. (selected but declined, unforeseen circumstances)
- Invited Delegate at the World Science Forum 2022, South Africa.
- Invited Delegate at the World Business Dialogue 2019, Cologne Germany.
- Jagriti Yatra 2016 India (selected but declined, unforeseen circumstances).
  §All India acceptance rate in 2016: ~ 3%

Peer-Reviewed International Journal Publications (Published/submitted: 29, In-prep: 4)

First-author work published [10, incl. corresponding: 2]

[**10**] *Organic Solvents-Based Offline Aerosol Mass Spectrometry (SOff-AMS) for Comprehensive Chemical Characterization of Ambient Organic Aerosol*
Khare, P.*, Ioannis, M.M., Dada, L., Kurdieh, A.A., Hao, Y., Diapouli, E., Voiron, C., Jaffrezo, J-L., Styszko, K., Skiba, A., Prevot, A.S.H., Daellenbach, K.R.
In **Environmental Science & Technology (American Chemical Society),** 59, 34, 18236–18248 (2025) **(I.F. 11.3)**
https://doi.org/10.1021/acs.est.5c08949

[**9**] *Chemical Composition of Fresh and Aged Asphalt-Related Organic Aerosols: From Ambient Observations to Laboratory Experiments*
Khare, P., Machesky, J., Williams, L., Humes, M., Canagaratna, M., Krechmer, J.E., Lambe, A.T., Presto, A.A., Gentner, D.R.
In **Environmental Science & Technology | Air (American Chemical Society)**, 2, 4, 446-455 (2025) **(I.F. pending)**
https://doi.org/10.1021/acsestair.4c00193

[**8**] *Detailed Chemical Characterization and Source Apportionment of Coarse Airborne Particulate Matter in Urban and Rural Catalonia, Spain*
joint-first-[in 't Veld**,** M., **Khare, P.**], Hao, Y., Reche, C., Pérez, N., Alastuey, A., Yus-Díez, J., Marchand, N., Prevot, A.S.H., Querol, X., Daellenbach, K.R.
In **Science of the Total Environment (Elsevier)**, 902, 166440 (2023) **(I.F. 8.0)**
https://doi.org/10.1016/j.scitotenv.2023.166440

[**7**] *Ammonium-adduct Chemical Ionization to Investigate Anthropogenic Oxygenated Gas-Phase Organic Compounds in Urban Air*
Khare, P., Krechmer, J.E., Machesky, J. E., Hass-Mitchell,T., Cao,C., Wang, J., Majluf, F., Lopez-Hilfiker, F., Seltzer, K., Pye, H.O.T., Commane, R., McDonald, B., Toledo-Crow, R., Mak, J.E., Gentner, D.R.
In **Atmospheric Chemistry and Physics (European Geosciences Union)**, 22, pp. 14377-14399 (2022) **(I.F. 5.1)**
https://doi.org/10.5194/acp-22-14377-2022

[**6**] *Asphalt-related Emissions are a Major Missing Non-Traditional Source of Secondary Organic Aerosol Precursors*
Khare, P., Machesky, J., Soto, R., He, M., Presto, A., Gentner, D.R.
In **Science Advances (Science/AAAS)**, 6(36), eabb9785 (2020) **(I.F. 12.5)**
https://doi.org/10.1126/sciadv.abb9785
This publication was covered by many popular media outlets worldwide: The Guardian, CNBC, NewScientist, Associated Press, MSN News, ABC News, Smithsonian Magazine, Phys.org among others.
Science| Research Highlight: "It's not just cars that make pollution. It's the roads they drive on, too." September 3, 2020.

[**5**] *Considering the Future of Anthropogenic Gas-Phase Organic Compound Emissions and the Increasing Influence of Non-Combustion Sources on Urban Air Quality.*
Khare, P., Gentner, D.R.
In **Atmospheric Chemistry and Physics (European Geosciences Union)**, 18, pp. 5391-5413 (2018) **(I.F. 5.1)**
https://doi.org/10.5194/acp-18-5391-2018
Selected as highlight article by the journal.

[4] *Advances in Offline Approaches for Trace Measurements of Complex Organic Compound Mixtures via Soft Ionization and High-Resolution Tandem Mass Spectrometry*
Khare, P., Marcotte, A., Sheu, R., Walsh, A., Ditto, J., Gentner, D.R.
In **Journal of Chromatography A (Elsevier)**, 1598, pp. 163-174 (2019) **(I.F. 4.0)**
https://doi.org/10.1016/j.chroma.2019.03.037

[**3**] *A Large Scale Investigation of the Quality of Groundwater in Six Major Districts of Central India during the 2010-2011 Sampling Campaign.*
Khare, P.*
In **Environmental Monitoring and Assessment (Springer International)**, 189:429. (2017) **(I.F. 3.0)**
https://doi.org/10.1007/s10661-017-6130-0

[**2**] *Simulation of Vertical Concentration Gradient of Influenza Viruses in Dust Resuspended by Walking.*
Khare, P., Marr, L.C.
In **Indoor Air (Wiley)**. Vol. 25, 428-440. (2014) **(I.F. 4.3)**
https://doi.org/10.1111/ina.12156

[1] *Droplet Spectrum Analysis from Artificially Generated Rain Showers.*
Khare, P., Agarwal, T., Ghosh, S.
In **ASCE Journal of Irrigation and Drainage Engineering**, Vol. 140(8), ISSN (online): 1943-4774. (2014) **(I.F. 2.1)**
https://doi.org/10.1061/(ASCE)IR.1943-4774.0000742

Co-authored papers published/submitted [19]

[19] *Molecular-Level Characterization and Identification of Sources of Ambient Organic Aerosol in Ho Chi Minh City*
Kurdieh et al.
(submitted)

[18] *Assessing Organic Aerosol Sources in Barcelona via Non-Targeted Molecular Characterization*
Arianna Tronconi et al
(submitted)

[17] *Regional and Urban Drivers of PM2.5 Oxidative Potential Across China*
Cui et al.
(submitted)

[16] *A Decadal-Scale Perspective on PM10 Composition and its Variability Drivers at the High-Alpine research station Jungfraujoch*
Julian Weng, Yufang Hao, Benjamin T. Brem, Tianqu Cui, **Peeyush Khare**, Lubna Dada, Kaspar R. Daellenbach, Sophie Darfeuil, Gaëlle Uzu, Jean-Luc Jaffrezo, Martin Steinbacher, Stefan Reimann, Thaleia Gkraikou, Konstantina Oikonomou, Jean Sciare, Martine Collaud Coen, Claudia Mohr, Andre S. H. Prévôt, Martin Gysel-Beer, Imad El Haddad, Patrik Winiger
In **Atmospheric Chemistry and Physics (European Geosciences Union)**
(accepted)

[15] *Recreating the Breath of Cities in the Atmospheric Simulation Chamber SAPHIR*
Yizhen Wu, Ralf Tillmann, Eva Y. Pfannerstill, **Peeyush Khare** + 60 co-authors + Georgios I. Gkatzelis.
In **Environmental Science & Technology (American Chemical Society)** (2026) 60 (42): 29992–30007 **(I.F. 12.2)**
https://doi.org/10.1021/acs.est.6c05767

[14] *Multi-Stress Interaction Effects on BVOC emission Fingerprints in Oak and Beech: A Cross-Investigation Using Neural Networks and Positive Matrix Factorization*
Biplob Dey, Toke Due Sjøgren, **Peeyush Khare**, Yizhen Wu, Sindhu Vasireddy, Georgios Gkatzelis, Martin Schultz, Alexander Knohl, Riikka Rinnan, Thorsten Hohaus, Eva Y. Pfannerstill
In **Biogeosciences (European Geosciences Union)** 23(4) 1423-1457 (2026) **(I.F. 3.9)**
https://doi.org/10.5194/bg-23-1423-2026

[13] *Real-Time Source Apportionment Approaches in Six Chinese Cities Using the AXA (ACSM, Xact, Aethalometer) Instrumental Set-Up*
Manousos I. Manousakas, Tianqu Cui, Qiyuan Wang, Lu Qi, Cheung Ka Yuen, Lubna Dada, Yufang Hao, **Peeyush Khare**, Tian Jie, Yuemei Han, Jay G. Slowik, Junji Cao, Kaspar R. Daellenbach, André S. H. Prévôt
In **Scientific Reports (Springer Nature)** 16 (9890) (2026) **(I.F. 3.9)**
https://doi.org/10.1038/s41598-026-38154-x

[12] *Characterization of Coarse Particulate Matter in Urban & Rural Switzerland using Advanced Offline Mass Spectrometry*
Ortiz Beltran K.S., Cui, T., Casotto, R., Lamkaddam, H., Tobler, A., Hao, Y., **Khare, P.**, Manousakas, M., Dada, L., Grange, S.K., Huglin, C., Uzu, G., Jaffrezo, J-L., Rausch, J., Jaramillo-Vogel, D., El-Haddad, I., Slowik, J.G., Prevot, A.S.H., Daellenbach, K.R.
In **Atmosphere (MDPI)** 17 (199) (2026) **(I.F. 2.3)**
https://doi.org/10.3390/atmos17020199

[11] *Assessing the Severe Urban Pollution Crisis in Sarajevo, Bosnia and Herzegovina: Mobile Measurements and Source Characterization*
Michael Bauer, Jay G. Slowik, Marta Via, **Peeyush Khare**, Benjamin Chazeaua, Kristina Glojek, Manousos Manousakas, Zachary C.J. Decker, Asta Gregorič, Almir Bijedić, Enis Krečinić, Griša Močnik, Katja Džepina, André S.H. Prévôt
In **Environment International (Elsevier)** 207 (110009) (2026) **(I.F. 9.7)**
https://doi.org/10.1016/j.envint.2025.110009
This publication was covered by several popular media outlets worldwide: The Guardian, Phys.Org among others.

[10] *Transported Smoke from Crop Residue Burning as the Major Source of Organic Aerosol and Health Risks in Northern Indian Cities During Post-Monsoon*
Hao, Y., Straehl, J., **Khare, P.**, Cui, T., Schneider-Beltran, K.S., Qi, L., Wang, D., Top, J., Surdu, M., Bhattu, D., Bhowmik, H.S., Vats, P., Rai, P., Kumar, V., Ganguly, D., Szidat, S., Uzu, G., Jaffrezo, J-L., Rastogi, N., Slowik, J., Haddad, I.E., Tripathi, S.N., Prevot, A.S.H., Daellenbach, K.R.
In **Environment International (Elsevier)** 202 (109583) (2025) **(I.F. 9.7)**
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[1] 10.5281/zenodo.16880946 - Mass spectral datasets for newly developed offline aerosol mass spectrometry technique. Associated paper: Organic Solvents-Based Offline Aerosol Mass Spectrometry (SOff-AMS) for Comprehensive Chemical Characterization of Ambient Organic Aerosol

Contributions in International Conferences/Meetings/Workshops

Corresponding author/presentor contributions [29 of 85 total]

Oral presentations [18]

[18] *Molecular Drivers of Organic Aerosol in Ambient Reactive Mixtures from Emerging Urban Sources Simulated in the SAPHIR Chamber*
International Aerosol Conference 2026, Xi'an, China
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[17] *Implementation Highlights from the Contributing Units to the in-situ Reactive Trace Gases (CiGas) Measurements in the ACTRIS Research Infrastructure Framework*
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[16] *Detailed Molecular Characterization of Limonene Secondary Organic Aerosol Under Varying Oxidation Conditions and Reaction Timescales at the SAPHIR Chamber during CHANEL campaign*
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[15] *Perspectives on External Quality Assessment of the VOC Data Reported by ACTRIS Measurement Platforms - Current Methods and Looking Forward*
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[14] *Molecular-Level Characterization of Secondary Organic Aerosol from Limonene Emissions Under Varying Oxidation Conditions and Reaction Timescales in the SAPHIR Chamber during CHANEL campaign*
Helmholtz Topic Days 2024: Changing Earth - Sustaining our Future, Bad Schönborn, Germany

[13] *Development of Solvents-Based Offline Aerosol Mass Spectrometry for Comprehensive Chemical Characterization of Ambient Organic Aerosol*
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[12] *Sources and Oxidation Potential of Fine Airborne Particulate Matter in Megacities of North and South India*
Molecular-Level Understanding of Atmospheric Aerosols (MUOAA) Meeting 2024, Corsica, France

[11] *Application of Offline Aerosol Mass Spectrometry for Identification and Characterization of Sources of Airborne Particulate Matter in Lucknow and Pune and their Oxidative Potential*
The Swiss Agency for Development and Cooperation (SDC)-India International Collaboration Meeting 2024, Villigen, Aargau Switzerland

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[11] *Demonstration of the Interoperability of Two Atmospheric Simulation Chambers*
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[10] *Constraining the Molecular-Level Chemical Composition and Sources of Air Pollution in Indian Megacities*
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782.Wilhelm und Else Heraeus-Foundation Seminar 2023 on Aerosols, Health and Climate: Gigacity and Future, Bad Honnef, Germany

[7] *Detailed Chemical Speciation and Source Apportionment of PM2.5 in Densely Populated Indo-Gangetic Region Using High-Resolution Mass Spectrometry*
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American Geophysical Union (2019), San Francisco USA

[5] *Emissions of Complex Gas-Phase Organic Compound Mixtures from the Application and Use of Asphalt Materials Under Typical Environmental Stresses and Emissions Timescales*
Gordon Research Conference 2019: Atmospheric Chemistry, Newry ME USA

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American Association for Aerosol Research (AAAR) - 10th International Aerosol Conference 2018, St. Louis MO USA

[3] *Next Generation Offline Approaches to Trace Organic Compound Speciation: Approaching Comprehensive Speciation with Soft Ionization and Very High Resolution Tandem Mass Spectrometry.*
American Geophysical Union (AGU) 2017, New Orleans LA USA

[2] *Evolving Urban Emissions: Increasing Impact of Non-Combustion Secondary Organic Aerosol Precursor Sources on Urban Air Quality.*
American Association for Aerosol Research Conference (AAAR) 2017, Raleigh NC USA

[1] *Stop Measuring Happiness from Material Gains.*
Leaders of Tomorrow Summit, St. Gallen Symposium 2016, Switzerland.

Co-authored contributions [56 of 85 total]

Please click here for a detailed list: Peeyush Khare - list of publications

Select Non-Technical Skills

Black Belt in Shotokan Karate (Conferred by Japan Karate Association)
Assistant Instructor in Krav Maga

References

Upon request

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